

DATA PREPROCESSING REPORT

Project : Advanced Regression Model Comparison & Optimization Platform

Dataset : House Prices - Advanced Regression Techniques

1. Objective

Data preprocessing is an essential step in machine learning. The objective of preprocessing is to improve data quality by handling missing values, removing inconsistencies, treating outliers, encoding categorical variables and scaling numerical features before model development.

2. Preprocessing Summary

Operation	Status
Missing Values	Handled
Duplicate Records	Removed
Outlier Treatment	Completed
Categorical Encoding	Completed
Feature Scaling	Applied
Skewness Reduction	Applied
Feature Selection	Completed

3. Missing Value Handling

Missing values were identified during data exploration. Numerical features were imputed using the median, while categorical features were filled using the most frequent value or an appropriate category where necessary.

Observation: After preprocessing, all missing values were successfully handled, ensuring a complete dataset for model training.

4. Outlier Treatment

Extreme values were identified using box plots and statistical analysis. Appropriate outlier treatment techniques were applied to reduce their impact while preserving meaningful observations.

5. Encoding and Feature Scaling

Categorical variables were converted into numerical form using encoding techniques. Numerical features were standardized using StandardScaler to ensure consistent feature scales and improve regression model performance.

6. Feature Engineering

Important features were selected based on correlation analysis and domain knowledge. Log transformation was applied to reduce skewness in highly skewed numerical features, resulting in a more balanced data distribution.

7. Key Improvements

- Missing values successfully handled.
- Duplicate records removed.
- Outliers treated to reduce noise.
- Categorical features encoded.
- Numerical features scaled.
- Skewed features transformed using logarithmic scaling.
- Final dataset prepared for regression model development.

8. Conclusion

Data preprocessing significantly improved the quality and consistency of the dataset. By addressing missing values, outliers, encoding, scaling and feature engineering, the dataset became suitable for training accurate and reliable regression models for house price prediction.